Transaction

- memberID : String
- memberName : String
- bookID : int
- bookTitle : String
- action : String - time : LocalDateTime
- dueDate : LocalDate
- fine : doubl

+ Transaction(member : Member, book : Book, action : String, dueDate : LocalDate, fine : double)
+ Transaction(time : LocalDateTime, memberID : String, memberName : String, bookID : int, bookTitle : String, action : String, dueDate : LocalDate, fine : double)
+ isBorrow() : boolean
+ getMemberID() : String
+ getMemberName() : String
+ getBookID() : int
+ getBookTitle() : String
+ getAction() : String
+ getTime() : LocalDateTime
+ getDueDate() : LocalDate
+ getFine() : double
+ getTimeText() : String
+ getDueText() : String
+ getFineText() : Strin

Library

+ MIN_USERNAME : int
+ MIN_PASSWORD : int
+ LOAN_DAYS : int
+ FINE_PER_DAY : double
- allBooks : ArrayList
- allMembers : ArrayList
- accounts : ArrayList
- transactions : ArrayList

+ load() : void
+ saveBooks() : void
+ saveUsers() : void
+ usernameProblem(username : String) : String
+ passwordProblem(password : String) : String
+ daysLate(dueDate : LocalDate) : long
+ fineFor(dueDate : LocalDate) : double
+ dateText(date : LocalDate) : String
+ moneyText(amount : double) : String
+ addBook(book : Book) : void
+ removeBook(book : Book) : void
+ findBook(id : int) : Book
+ bookExists(title : String, author : String, category : String, year : int, ignoreID : int) : boolean
+ nextBookID() : int + getBooks() : ArrayList
+ addMember(member : Member) : void
+ removeMember(member : Member) : void
+ findMember(id : String) : Member
+ getMembers() : ArrayList
+ login(username : String, password : String) : User
+ usernameExists(username : String) : boolean
+ borrowBook(member : Member, book : Book) : boolean
+ returnBook(member : Member, book : Book) : boolean
- record(t : Transaction) : void
+ getTransactions() : ArrayList
+ getTransactionsOf(memberID : String) : ArrayList

Book

- author : String
- category : String - year : int
- cover : String
- totalStock : int
- availableStock : int

+ Book(itemID : int, title : String, author : String, category : String, year : int, totalStock : int)
+ setAuthor(author : String) : void + setCategory(category : String) : void
+ setYear(year : int) : void
+ restoreAvailableStock(available : int) : void
+ setStock(newTotal : int) : boolean + displayDetails() : void
+ borrowBook() : boolean
+ returnBook() : boolean
+ getAuthor() : String
+ getCategory() : String
+ getYear() : int
+ getStock() : int
+ getAvailableStock() : int
+ getBorrowedCount() : int
+ getCover() : String
+ setCover(cover : String) : void

<<abstract>>
LibraryItem

- itemID : int - title : String
- isAvailable : boolean

+ LibraryItem(itemID : int, title : String, isAvailable : boolean)
+ setTitle(title : String) : void
+ displayDetails() : void
+ getItemID() : int
+ getTitle() : String
+ isAvailable() : boolean
+ setAvailability(isAvailable : boolean) : void

FileManager

- DIR : String
- BOOKS : String
- USERS : String
- TRANSACTIONS : String
- TIME_FORMAT : DateTimeFormatter - SPLIT : String

+ booksFileExists() : boolean
+ usersFileExists() : boolean
- clean(text : String) : String
- value(field : String, label : String) : String
- skip(line : String) : boolean
- writeHeader(writer : BufferedWriter, lines : String...) : void
- openReader(path : String) : BufferedReader
- openWriter(path : String, append : boolean) : BufferedWriter
+ readBooks() : ArrayList
+ writeBooks(books : ArrayList) : void
+ readUsers() : ArrayList
+ writeUsers(users : ArrayList) : void
- parseTime(text : String) : LocalDateTime
- writeTransactionHeader(writer : BufferedWriter) : void
- transactionLine(t : Transaction) : String
+ readTransactions() : ArrayList
+ writeTransactions(list : ArrayList) : void
+ appendTransaction(t : Transaction) : void

<<abstract>>
User

- userID : String
- name : String
- username : String
- password : String

+ User(userID : String, name : String, username : String, password : String)
+ getRole() : String
+ getUserID() : String + getName() : String
+ getUsername() : String
+ setUsername(username : String) : void
+ getPassword() : String
+ setPassword(password : String) : void
+ checkPassword(password : String) : boolean
+ checkLogin(username : String, password : String) : boolean
+ displayDetails() : void

Member

- borrowedBooks : ArrayList
- dueDates : HashMap

+ Member(userID : String, name : String, username : String, password : String)
+ displayDetails() : void
+ borrowBook(book : Book) : boolean
+ returnBook(book : Book) : boolean
+ restoreBorrowed(book : Book, dueDate : LocalDate) : void
+ releaseBorrowed(book : Book) : void
+ getDueDate(book : Book) : LocalDate
+ getRole() : String + getBooks() : ArrayList

Frames

username : JTextField
password : JPasswordField
whitepanel : JPanel
greenpanel : JPanel
loginbtn : JButton
confirmbtn : JButton
userlbl : JLabel
passlbl : JLabel
confirmlbl : JLabel
Lib : JLabel

+ Frames()

LoginPage

- library : Library

+ LoginPage(library : Library)
+ actionPerformed(e : ActionEvent) : void
+ login() : void
+ showLogin() : void

MemberGUI

- lb : Library
- member : Member
- onLogout : Runnable
- updatingFilters : boolean = false
~ userPanel : JPanel
~ Bookpanel : JPanel
~ line1 : JPanel
~ line2 : JPanel
~ line3 : JPanel
~ line4 : JPanel
~ line5 : JPanel
~ welcomeLabel : JLabel = "Welcome, Member"
~ subtitleLabel : JLabel = "Greenfield University Library"
~ previewHeader : JLabel = "Book Preview"
~ PictureLabel : JLabel = "Book Cover"
~ TitleLabel : JLabel = "Title:"
~ AuthorLabel : JLabel = "Author:"
~ CategoryLabel : JLabel = "Category:"
~ YearLabel : JLabel = "Pub. Year:"
~ AvailabilityLabel : JLabel = "Status:"
~ logo : JLabel
~ borrowedCountLabel : JLabel
~ column : String[5] = {"ID", "Title", "Author", "Category", "Status"}
~ model : DefaultTableModel
~ table : JTable
~ scrollPane : JScrollPane
~ center : DefaultTableCellRenderer
~ txtSearch : JTextField = "Search title or author"
~ Category : JComboBox<String>
~ statusBox : JComboBox<String>
borrowBtn : JButton = "Borrow a Book"
retrunBtn : JButton = "Return a Book"

+ MemberGUI(Library library, Member member, Runnable onLogout)
+ membertableFormat() : void
+ refreshTable() : void
+ refreshCategories() : void
+ applyFilters() : void
- getSelectedBook() : Book
- showDetails(Book b) : void
- clearDetails() : void
- openBorrowReturn(boolean borrowing) : void
- showRecords() : void
- showProfile() : void
- logout() : void
+ setHintPlaceholder(JTextField textField, String hint) : void
+ getFieldText(JTextField textField, String hint) : String

Librarian

+ Librarian(userID : String, name : String, username : String, password : String)
+ addBook(library : Library, book : Book) : void
+ removeBook(library : Library, bookID : int) : boolean
+ addMember(library : Library, member : Member) : void
+ getRole() : String

«Interface»
MemberGUI

- lb: Library
- member: Member
- onLogout: Runnable
- updatingFilters: boolean

+ MemberGUI(Library: library, Member: member, Runnable: onLogout)
+ membertableFormat : void
+ refreshTable() : void
+ applyFilters() : void
- getSelectedBook() : void
- showDetails(Book: b) : void
- clearDetails() : void
- openBorrowReturn(boolean: borrowing) : void
- showRecords() : void
- showProfile() : void
- logout() : void
+ setHintPlaceholder(JTextField: textField, String: hint) : void
+ getFieldText(JTextField: textField, String: hint) : String

Main

+ main(args : String[]) : void

«Interface»
TransactionDialog

.

+ TransactionDialog(JFrame: owner, String: heading, ArrayList<Transaction>: list, boolean: showMember)
+ smoothIcon(String: path, int: maxW, int: maxH) : Icon

LoginPage

- library : Library

+ LoginPage(Library library)
+ login() : void
+ showLogin() : void

h

- hovered: boolean

+ IconButton(String: iconPath, String: caption, Runnable: action)
+ place(int: x, int: y)
paintComponent(Graphics: g) : void

LibrarianGUI

- lb : Library
- librarian : Librarian
- onLogout : Runnable
- imagePath : String
- column : String
- options : String[]
- selectedBookID : int
- updatingFilters : boolean
~ model : DefaultTableModel
~ logo : ImageIcon
~ table : JTable
~ scrollPane : JScrollPane
~ addStock : JSpinner
~ center : DefaultTableCellRenderer
~ cbStatus : JComboBox
~ cbCategory : JComboBox
~ cbYear : JComboBox
~ addBookPanel : JPanel
~ detailsPanel : JPanel
~ userPanel : JPanel
~ titleLabel : JLabel
~ authorLabel : JLabel
~ categoryLabel : JLabel
~ yearLabel : JLabel
~ stockLabel : JLabel
~ pTitleLabel : JLabel
~ pAuthorLabel : JLabel
~ pCategoryLabel : JLabel
~ pPubYearLabel : JLabel
~ pStockLabel : JLabel
~ welcomeLabel : JLabel
~ pPictureLabel : JLabel
~ cPictureLabel : JLabel
~ logoLabel : JLabel
~ txtTitle : JTextField
~ txtAuthor : JTextField
~ txtCategory : JTextField
~ txtYear : JTextField
~ txtSearch : JTextField
~ addBookBtn : JButton
~ chooseCoverBtn : JButton
~ clearBtn : JButton
~ removeBookBtn : JButton
~ editBookBtn : JButton
~ cancelEditBtn : JButton

+ LibrarianGUI(library, Library, librarian Librarian, onLogout Runnable)
+ userPanelFormat() : void
+ refreshTable() : void
+ refreshCategories() : void
- rowOf(b : Book) : Object[]
+ tableFormat() : void
+ addBookFormat() : void
+ detailsPanelFormat() : void
+ searchActionPerformed() : void
+ setHintPlaceholder(textField : JTextField, hint : String) : void
+ getFieldText(textField : JTextField, hint : String) : String
+ applyFilters() : void
+ chooseCoverActionPerformed() : void
+ clearBtnActionPerformed() : void
+ addBtnActionPerformed() : void
+ editBtnActionPerformed() : void
+ removeBtnActionPerformed() : void
+ loadSelectedBook() : void
+ finishEditing() : void
- sameCover(a : String, b : String) : boolean
- onSelectionChanged() : void
- showDetails(b : Book) : void
- clearDetails() : void
- showTransactions() : void
- showMembers() : void
- showProfile() : void
- logout() : void
+ actionPerformed(e : ActionEvent) : void

«Interface»
LibrarianGUI

- lb: Library
- imagePath: String
- column: String

+ LibrarianGUI(Library: library, Librarian: librarian, Runnable: onLogout)
+ userPanelFormat() : void
+ refreshCategories() : void
+ loadSampleData() : void
+ refreshTable() : void
+ tableFormat() : void
+ addBookFormat() : void
+ detailsPanelFormat() : void
+ searchFormat() : void
+ setHintPlaceholder(JTextField: textField, String: hint)
+ chooseCoverActionPerformed : void
+ applyFilters : void
+ getFieldText(JTextField: textField, String: hint) : String
+ chooseCoverActionPerformed() : void
+ clearBtnActionPerformed() : void
+ addBtnActionPerformed() : void
+ editBtnActionPerformed() : void
+ removeBtnActionPerformed() : void
+ loadSelectedBook() : void
+ finishEditing() : void
- sameCover(String: a, String: b) : boolean
- onSelectionChanged() : void
- showDetails(Book b) : void
- clearDetails() : void
- showTransactions() : void
- showMembers() : void
- showProfile() : void
- logout() : void

«Interface»
ProfileDialog

- lb: Library
- user: User
- usernameField: JTextField

+ ProfileDialog(JFrame: owner, Library: lb, User: user)
- addRow(String text, int y) : void
- styleReadOnly(JTextField: field) : void
- smallButton(String text) : JButton
- editUsername() : void
- changePassword() : void
- warn(String message) : void

Main

+ main(args : String[]) : void

«BorrowReturnDialog»
Interface

- Library lb
- Member member
- boolean borrowing
- String[] borrowColumns
- String[] returnColumns
- DefaultTableModel model
- Jtable table
- JTextField txtSearch
- JLabel infoLabel
- JButton actionBtn

«Interface»
ImageUtil

.

+ scale(String: path, int: maxW, int: maxH) : ImageIcon
+ smoothIcon(String: path, int: maxW, int: maxH) : Icon

«Interface»
BorrowReturnDialog

- lb: Library
- member: Member
- borrowing: boolean
- table: JTable
- txtSearch: JTextField
- infoLabel: JLabel
- actionBtn: JButton

+ BorrowReturnDialog(JFrame: owner, Library: lb, Member: member, boolean)
- refreshList() : void
- doAction() : void

«Interface»
MemberDialog

- lb: Library
- admin: Librarian
- table: JTable

+ MembersDialog(JFrame owner, Library lb, Librarian admin)
- JButton makeButton(String: text, String: color, int: x)
- refresh() : void
- getSelectedMember() : Member
- addMember() : void
- changePassword() : void
- deleteMember() : void
- warn(String: message) : void